Upconversion Coating: Behavior in Detail

How an infrared-excited $NaYF_4$:Yb,Er/Tm coating behaves

1. Diffuse glow, not a mirror reflection

- **Emission is isotropic.** Each microcrystal (typically 0.1 to 10 µm) emits visible photons in random directions, so the coated spot looks like a small light source rather than a reflector. Brightness follows roughly a Lambertian (matte) pattern.
- **The binder and particles scatter light.** Powder in a binder is a turbid layer, so much visible light bounces around before escaping. A thicker layer gives more glow up to a point, then deeper layers receive too little excitation.
- **The IR beam reflects separately.** The 980 nm light bouncing off glass or the coating is specular but invisible. What you see is only the visible emission, which is unrelated to the mirror-like path of the beam.
- **Where the "metallic" look comes from.** On polished metal or a reflective backing, emitted light bounces back and the spot reads as a bright, hard-edged shine. A white or silver backing can nearly double visible output.
- **Color shifts with conditions.** Yb/Er materials glow redder relative to green at higher power density and temperature, so a spot looks greener at low power and more orange-yellow when driven hard.

2. Low efficiency

- **Typical numbers.** Bulk hexagonal $NaYF_4$:Yb,Er powder reaches roughly a few percent quantum yield at high power density. Nanoparticles and poorly made coatings are often well below 1%. Hexagonal phase is about ten times better than cubic phase, so phase purity matters.
- **It is a nonlinear process.** Two (sometimes three) IR photons are needed per visible photon, so at low power the output scales roughly with the square of power density. Doubling power gives about four times the brightness, saturating toward linear at high density.
- **Efficiency rises with power density.** The same laser converts far better when concentrated, which is why efficiencies are quoted at specific W/cm^2 values.
- **Quenching factors.** Water and hydroxyl groups on the surface, high dopant concentration, small particle size, and heating all steal energy before it becomes light. Poorly dried coatings can reduce output noticeably.
- **Visibility.** The light is dim and diffuse, so it is hard to see in bright rooms or sunlight, but a faint green spot is easy to see in a dim room because the eye is sensitive to green.

3. The beam needs to be focused

- **Power per area is what matters.** 100 mW spread over 1 cm^2 is only 0.1 W/cm^2, barely visible. The same 100 mW focused to 1 mm^2 is about 10 W/cm^2, which is bright. Because of the squared relationship, tightening the spot beats adding power.
- **Detector cards give a reference.** Commercial IR cards show a clear glow from a few milliwatts in a small spot. Wide, divergent LED beams usually do little.
- **Relevance to a window setup.** Each glass surface reflects about 4% at normal incidence, so a window returns roughly 4 to 8% of the beam. Because the response is nonlinear, the glow could be around 100 times dimmer than from the direct beam. To compensate: a stronger laser, a tighter focus on the target, a higher angle of incidence (which raises reflectivity), or coated glass.

- **Practical tip.** Keep the beam collimated until the window, then use a lens after the reflection to shrink the spot on the coating.

4. Emission persists for microseconds to milliseconds

- **Energy levels set the timing.** Ytterbium holds its excitation for about 1 ms, and the emitting erbium or thulium states live roughly 100 to 500 µs. The glow rises and fades over tens to hundreds of microseconds.
- **It acts as a low-pass filter.** Flicker faster than a few kHz is averaged out, so a PWM laser at tens of kHz looks steady. Pulsing below about 100 Hz passes through almost unchanged.
- **The eye already blurs flicker above 60 to 100 Hz,** so the dampening is mostly visible on a high-speed camera, a photodiode and oscilloscope, or with slow deliberate pulsing.
- **It is not glow-in-the-dark.** Persistent phosphors like strontium aluminate glow for minutes to hours; upconversion stops almost immediately by comparison.
- **Pulses behave differently.** A short, high-peak pulse yields much more visible light per unit of average power than a steady beam, with the lifetime spreading that burst out in time.

Safety: 980 nm light is invisible, so the dim glow can mislead you into thinking the beam is weak. Anything above a few milliwatts needs laser goggles rated for that wavelength, and stray reflections off a window are a real hazard.